

Analysis group meeting

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- Fix my issue with CLARA to test the multi thread, thanks to Rafaella and Nathan.

RG-D Vertex Studies Carbon: μ and σ values

	Vx		Vy	
	Mean [cm]	Sigma [cm]	Mean [cm]	Sigma [cm]
Sector independent	-0.00431889	0.138538	-0.175104	0.184344
Sector 1	0.00207391	0.0568253	-0.320168	0.249121
Sector 2	0.273244	0.270524	-0.27552	0.133421
Sector 3	0.0489401	0.213238	-0.154027	0.168174
Sector 4	-0.0290649	0.0975361	0.214474	0.283372
Sector 5	-0.250731	0.252091	-0.0754824	0.135787
Sector 6	-0.0224166	0.207086	-0.182258	0.17224

		Mean [cm]	Sigma [cm]
Vz	Entrance of the empty LD2 cell	-18.4237	1.16015
	Exit of the empty LD2 cell	-13.8489	1.02154
	first foil	-8.50419	0.691596
	second foil	-3.59758	0.815107

RG-D Vertex Studies CuSn: μ and σ values

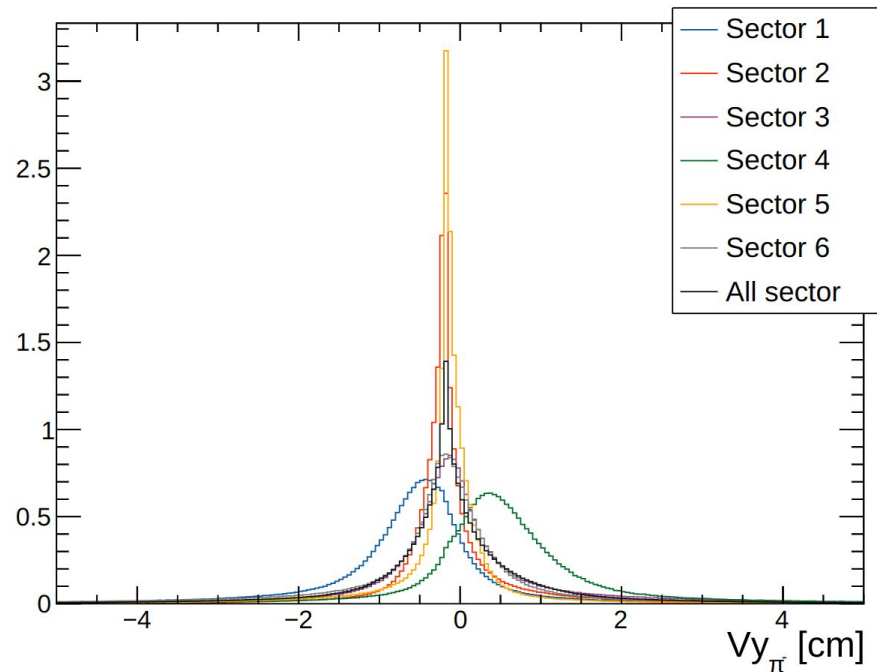
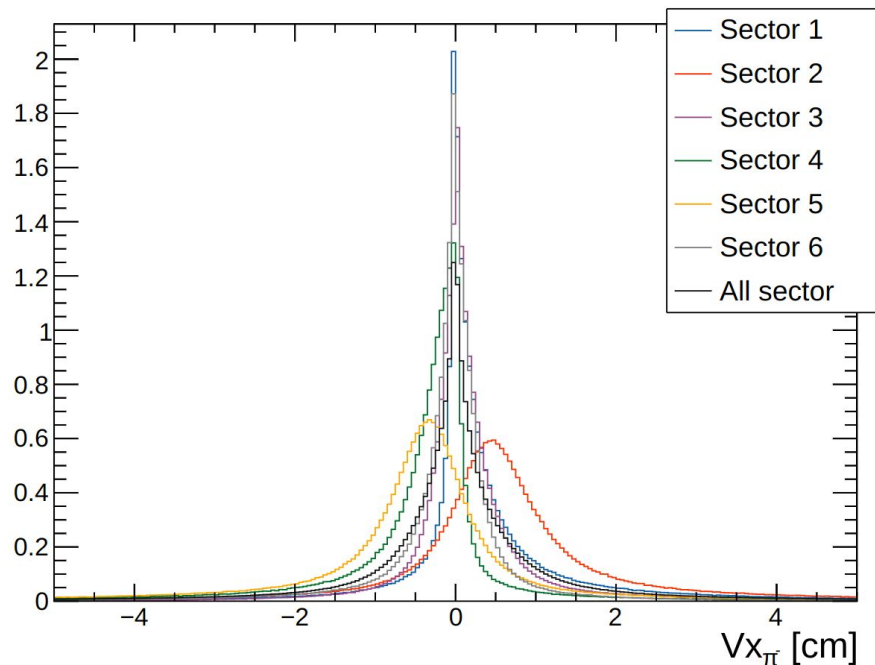
	Vx		Vy	
	Mean [cm]	Sigma [cm]	Mean [cm]	Sigma [cm]
Sector independent	-0.00421593	0.138723	-0.176477	0.185352
Sector 1	0.00104791	0.0560966	-0.316161	0.253364
Sector 2	0.270278	0.273344	-0.279131	0.133888
Sector 3	0.0482362	0.21678	-0.156601	0.170709
Sector 4	-0.0279068	0.0987972	0.22215	0.288263
Sector 5	-0.253408	0.253318	-0.0739325	0.137628
Sector 6	-0.0178941	0.210021	-0.180336	0.17197

		Mean [cm]	Sigma [cm]
Vz	Entrance of the empty LD2 cell	-18.4969	0.896509
	Exit of the empty LD2 cell	-13.803	0.779741
	Cu foil	-8.56379	0.662424
	Sn foil	-3.70048	0.812302

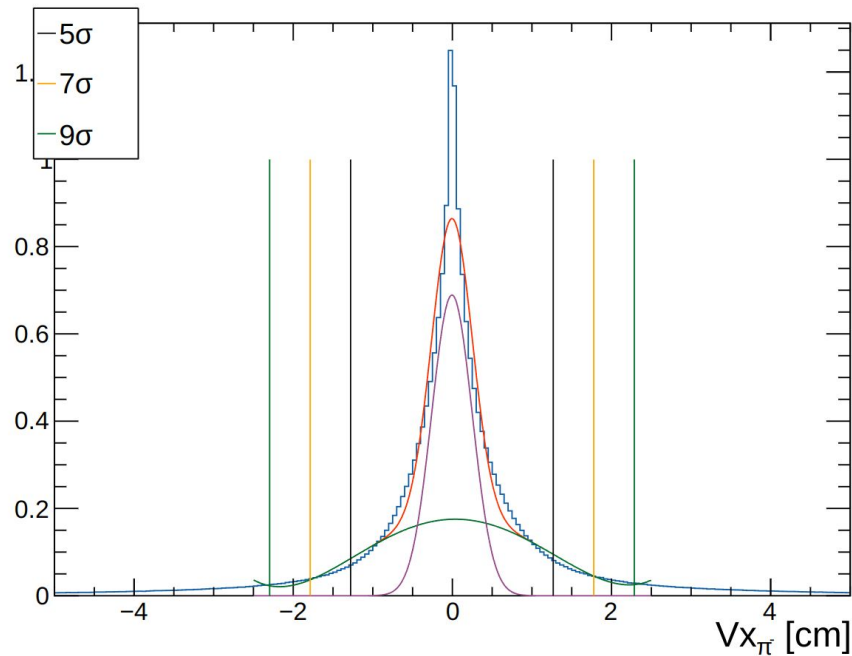
RG-D Vertex study: pion minus

Pions minus after Vx, Vy and Vz cuts

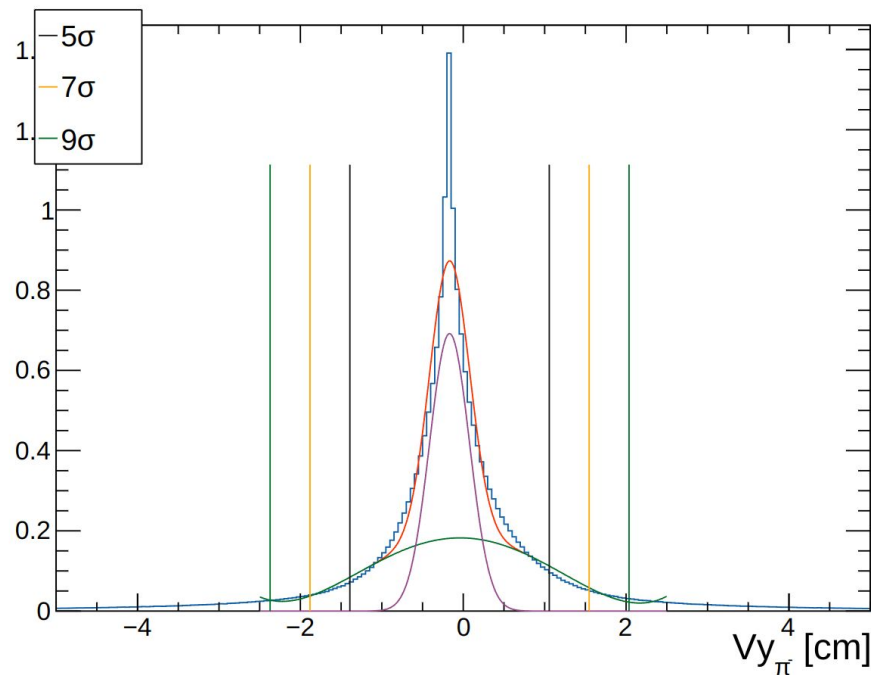
```
const double cut_vx_min = -0.00431889 - 11.0 * 0.138538;
const double cut_vx_max = -0.00431889 + 11.0 * 0.138538;
const double cut_vy_min = -0.175104 - 11.0 * 0.184344;
const double cut_vy_max = -0.175104 + 11.0 * 0.184344;
const double cut_vz_min = -8.50419 - 3 * 0.691596;
const double cut_vz_max = -3.59758 + 3 * 0.815107;
```



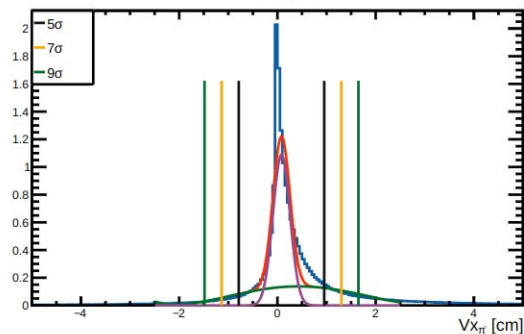
All Sector



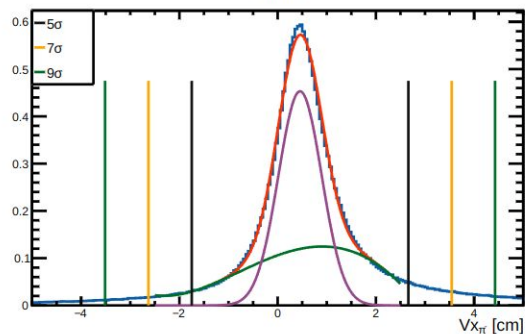
All Sector



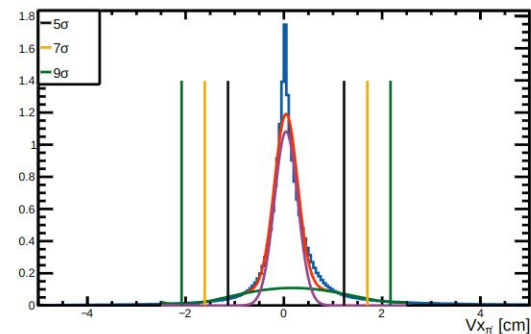
Sector 1



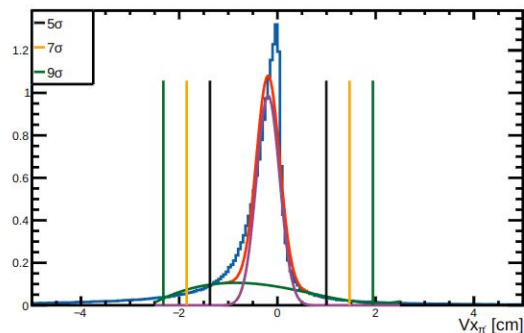
Sector 2



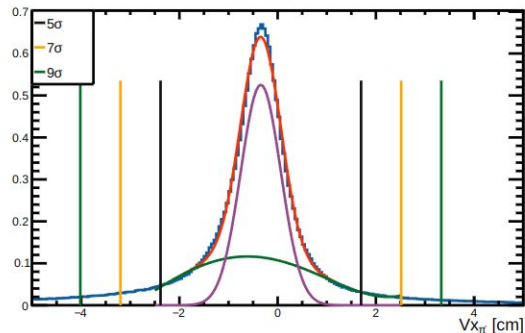
Sector 3



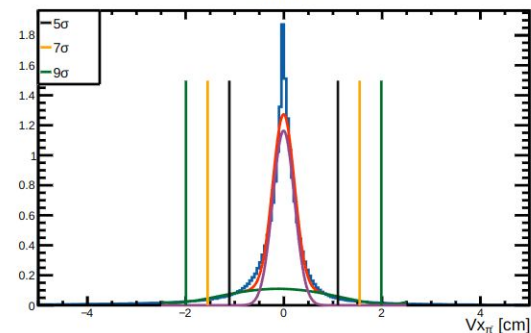
Sector 4



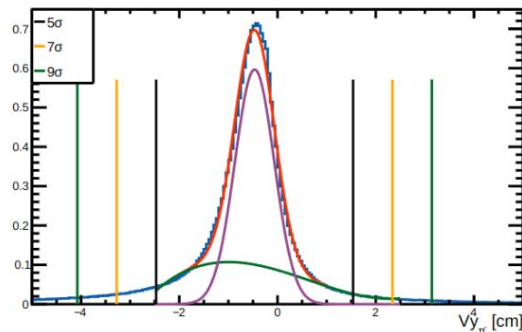
Sector 5



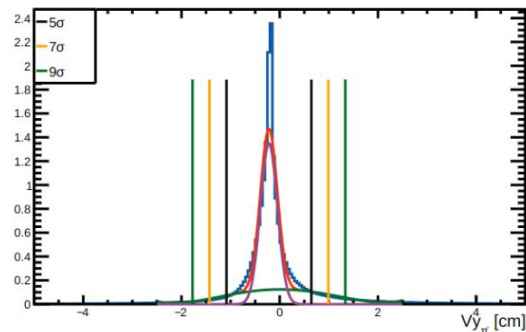
Sector 6



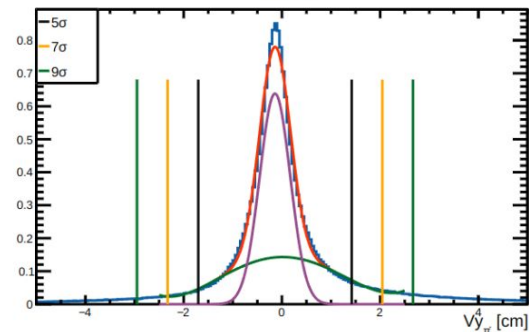
Sector 1



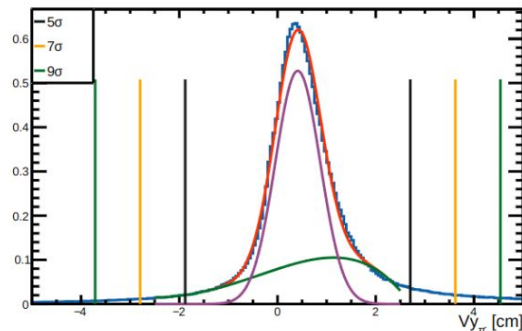
Sector 2



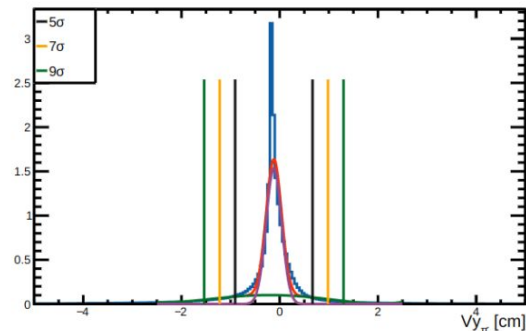
Sector 3



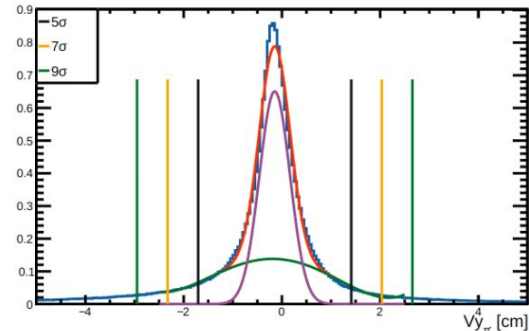
Sector 4

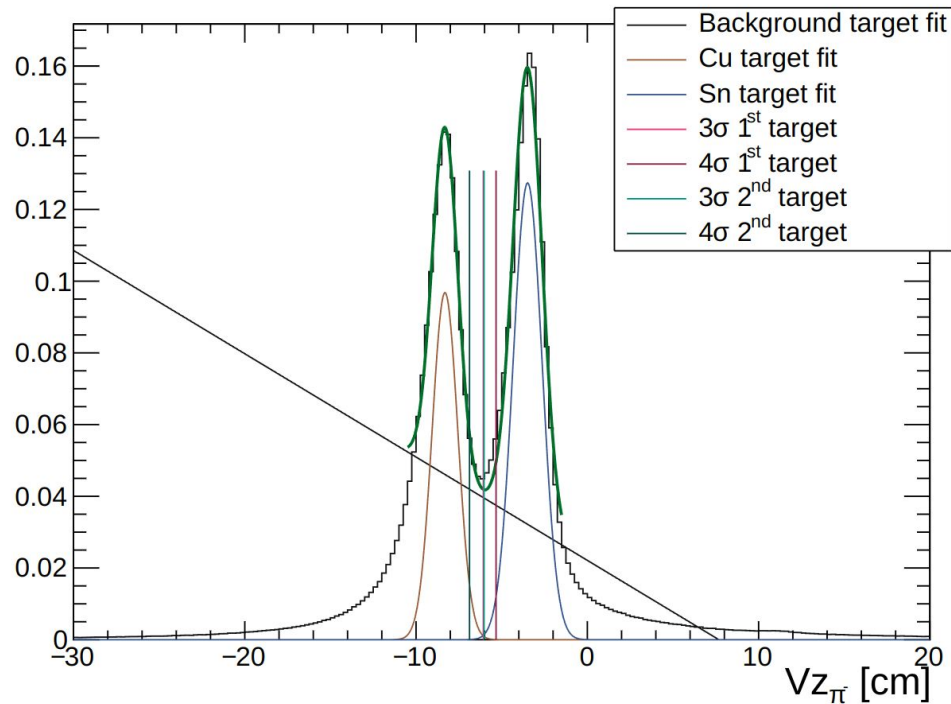
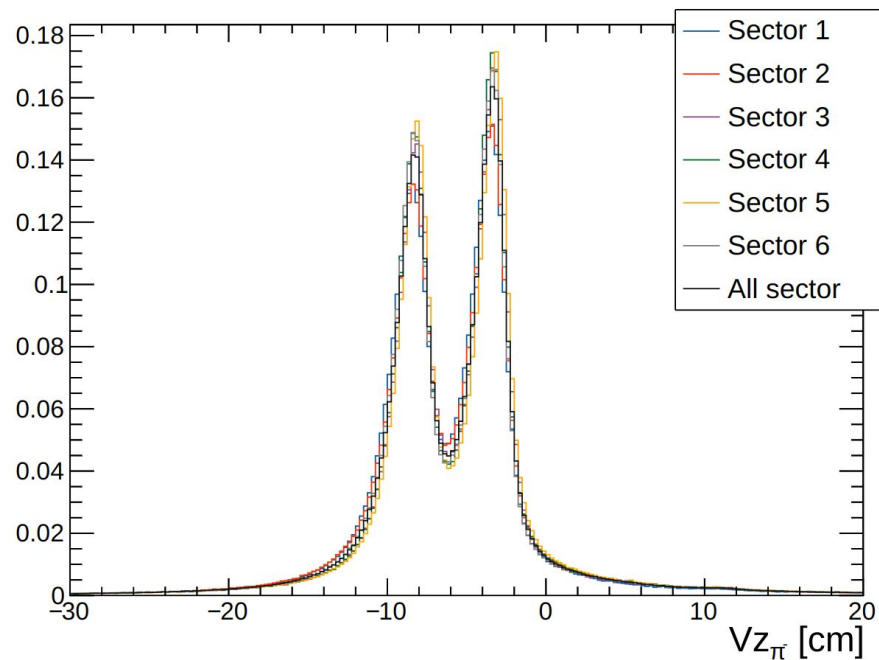


Sector 5



Sector 6





RG-D Vertex study: pion minus Carbon

	Vx		Vy	
	Mean [cm]	Sigma [cm]	Mean [cm]	Sigma [cm]
Sector independant	-0.00514765	0.254569	-0.167978	0.244907
Sector 1	0.0823251	0.174071	-0.466764	0.401006
Sector 2	0.456328	0.441058	-0.219773	0.172871
Sector 3	0.0449543	0.236303	-0.14104	0.312123
Sector 4	-0.189492	0.236972	0.412643	0.458341
Sector 5	-0.343068	0.408256	-0.119481	0.157631
Sector 6	-0.00472142	0.220994	-0.146407	0.311463

		Mean [cm]	Sigma [cm]
Vz	first foil	-8.31203	0.745773
	second foil	-3.48624	0.848342

- ALERT@22GeV
 - Generate more event.
 - Compute the errors with 50% of the total luminosity and 20 PAC days
 - $A_{LU}(90^\circ)$ as function of $-t$
- AI for ALERT:
 - Test the implement of the AI in COATJAVA in multi threaded (CLARA)
 - Recreate the efficiency/purity plot using COATJAVA
- RG-D study:
 - Vz sector depend fit
 - Pion plus/minus with Sn and Cu target